Science Review – Unit 2: Sound

Hải Phong · 4B3 · Tổng hợp từ LMS tuần 11–13 (26/10 – 13/11) · Name: _____ Date: _____

> Phụ huynh: Unit 2 (Sound) chưa có đề ôn trên LMS nên đây là phiếu tổng hợp từ bài giảng 2.1, 2.2, 2.3 của Mr Tim. Đáp án ở trang cuối.

1. Key ideas

2.1 How are sounds made?

Sounds are made when something **vibrates**. A **vibration** is a very quick movement backwards and forwards (you can feel it on your throat when you hum). You **pluck** a guitar string to make it vibrate.
Sound travels from the **source** to our ears as vibrations through a material. It travels **best through solids** (especially metals), then liquids, and **worst through air** – because particles in a solid are close together and pass the vibration on.
Sound gets **fainter** the further it travels from the source. Sound cannot travel through empty space (a vacuum).

2.2 Volume and pitch

Volume = how loud or quiet a sound is. **Bigger** vibrations → louder sound; smaller vibrations → quieter sound.
We measure volume with a **sound level meter**; the unit is the **decibel (dB)**.
Sounds louder than about **85 dB** can damage our ears.
Pitch = how high or low a sound is. **Fast** vibrations → **high-pitched** (a whistle, a bird); **slow** vibrations → **low-pitched** (thunder, a drum). Some sounds are too high (bats) or too low (elephants) for humans to hear.
On a guitar: a shorter, tighter or thinner string gives a higher pitch.

2.3 Changing the volume of sound

We can make a sound quieter by **muffling** it with soft materials that **absorb** sound (foam, cloth, carpet, cotton wool), by moving further from the source, or by blocking it (ear defenders, closed windows, walls, trees). Hard, smooth surfaces **reflect** sound and make echoes.
Noise pollution = unwanted or harmful loud sound. Ways to reduce it: turn the TV/music down, talk with an indoor voice, don't honk the horn, plant trees, wear ear defenders near loud machines.

Fair test words

Independent variable	Dependent variable	Control variables
the one thing you **change**	the thing you **observe / measure**	things you keep **the same**

e.g. the material around a buzzer	e.g. the volume in dB	e.g. same buzzer, same distance, same meter

2. Practice

1. All sounds are made by _____.
2. Does sound travel best through wood, water or air? _____ Why? _____
3. What happens to a sound as you walk away from it? _____
4. What do we use to measure the volume of a sound? _____ What is the unit? _____
5. Big vibrations make a _____ sound. Fast vibrations make a _____-pitched sound.
6. Write high or low pitch: a mouse squeaking _____; a lion roaring _____; a whistle _____; thunder _____.
7. Name one loud animal and one soft (quiet) animal. _____ / _____
8. Why can very loud sounds be dangerous? _____
9. Circle the materials that would be best to muffle an alarm clock: metal tray, cotton wool, glass, thick towel, foam.
10. Write three ways to reduce noise pollution at home or at school.
    a) _____ b) _____ c) _____
11. Zara tests which material makes a buzzer quietest. She wraps the buzzer in different materials and measures the sound with a sound level meter 1 metre away.
    Independent variable: _____ Dependent variable: _____
    One control variable: _____
12. Thanh plucks a short rubber band and a long rubber band. Which gives the higher pitch? _____ Why? _____

Answer key

1 vibrations (things vibrating) · 2 wood – it is a solid, its particles are close together so they pass the vibrations on well · 3 it becomes fainter/quieter · 4 a sound level meter; decibels (dB) · 5 loud; high · 6 high; low; high; low

7 e.g. loud: lion, dog, elephant; soft: cat, mouse, sloth, butterfly · 8 sounds above about 85 dB can damage our ears / hearing · 9 cotton wool, thick towel, foam · 10 e.g. turn the TV down, talk quietly, close the windows, plant trees, wear ear defenders, don't honk

11 independent: the material; dependent: the volume (dB); control: same buzzer / same distance (1 m) / same meter · 12 the short band – it vibrates faster, so the pitch is higher

Nguồn: LMS CIE Science 4 – Unit 2.1 How are sounds made (tuần 11), Unit 2.2 Volume and pitch (tuần 12), Unit 2.3 Changing the volume of sound + bài tập Noise pollution (tuần 13).